Harry Zhu

(647)-636-8871 | h46zhu@uwaterloo.ca | harryzhu.ca | GitHub | LinkedIn

Education

University of Waterloo — Waterloo, ON
Honours Bachelor of Computer Science (Data Science) — *Apr 2027*

- Courses: Data Structures, Object-Oriented Programming, Algorithm Design, Statistics, Linear Algebra, Calculus

Technical Skills

Languages: Java, Rust, C#, C/C++, TypeScript, Python, SQL, R
Frameworks: .NET, Spring Boot, React, Next.js, Blazor, Angular, Flask, Node.js, Pandas, NumPy, TensorFlow, SciKit
Developer Tools: SQL Server, PostgreSQL, Redis, Docker, Azure, Git, Maven, Gradle, Firebase, AWS, Vercel

Experience

Amazon Web Services — May 2025 – Present
Software Development Engineer — *Seattle, WA*

- Designed an auto re-enablement system for a **low-latency routing mechanism** in DynamoDB's routing layer, **increasing coverage from 73.5% to 90.8%** across all regions, enabling more customers to experience reduced latencies.
- Extended the mechanism to more data plane APIs, delivering a **750μs (~27.3%) reduction at P50** and **870μs (~15.8%) reduction at P99.9** in preliminary testing across SFO and IAD regions.
- Built a **burst-aware backoff model** using token bucket algorithm, applying jitter to **smooth retry traffic** and prevent overload under sustained high-throughput scenarios.
- Redesigned a **reverse bias back-off algorithm** to handle increasing replica counts, **ensuring future scalability** and maintaining traffic balance under high load.

TD Bank — Jan 2025 – Apr 2025
Software Developer — *Toronto, ON*

- Developed an **option pricing tool** with Python, leveraging NumPy, SciPy, and Bloomberg API to estimate the value of illiquid or unavailable options, saving analysts **10+ hours a week**.
- Integrated automated frameworks using Python and NumPy to **back-test quantitative investment strategies**, processing datasets of **10M+ data points**.
- Engineered a private debt capital call tool in React, Django, and PostgreSQL, handling **$68B+ in assets** to streamline capital call processes for portfolio managers.

BTNX — Apr 2024 – Aug 2024
Full-Stack Developer — *Toronto, ON*

- **Led the development of a new ERP SaaS product** targeted towards small to mid-sized distribution centers with Blazor and .NET along with designing the relational database in SQL Server.
- Optimized **SQL Server query speeds by 34%** by introducing a lookup Redis cache and migrating schema to use indexes
- Engineered a **price optimization API** using Python Flask and statsmodels, **increasing sales revenue by 4%** through dynamic pricing based on product and client.

Open Sails Partners — Jan 2024 – Aug 2024
Software Developer — *Toronto, ON*

- Developed an Express API with OCR tools to extract and analyze data from CIMs, **reducing data processing time by 70+%** and automating financial model construction.
- Implemented web scraping tools with Puppeteer to streamline deal sourcing, **improving lead sourcing time by 20%**.

Projects

Stock News Sentiment | *Python, TensorFlow, Pandas, Numpy, Sklearn*

- Developed a language processing model trained on 50,000+ financial articles to analyze the sentiment of financial text.
- Fine-tuned a Gated Recurrent Unit model, achieving **88% accuracy** across three distinct classes.

Hands-on Fluency | *JavaScript, Python, React, Flask, Tensorflow, OpenCV, MediaPipe, SQLite*

- Developed a sign language learning web platform with **React** and **Flask** that teaches and quizzes the user using their webcam with **OpenCV**.
- Leveraged Jupyter Notebook, Tensorflow, and MediaPipe to design and implement a **Convolutional Neural Network** for hand gesture recognition, resulting in a **71% accuracy across 30+ classes**.